

NGUYEN GIA HUY

FinTech Backend Engineer – Core Banking

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PROFESSIONAL SUMMARY

Core banking engineer at Vikki Digital Bank, building the **Lending** and **Overdraft** products on **Thought Machine Vault Core** (interest accrual, penalty and late fees, repayment, and settlement) together with the integrations, migration tooling, and test infrastructure around them. Nearly three years of experience, including fintech platform security at VNPAY Cloud (tenant isolation for **20+** fintech and enterprise customer organizations) and transaction integrity across **5+** blockchain networks.

TECHNICAL SKILLS

Core Banking: Thought Machine Vault Core, Python Smart Contracts, product simulation testing
Banking Products: Lending, overdraft; interest accrual, penalty and late fees, repayment, settlement
Languages: Python, Golang, SQL
Integration & Data: gRPC, REST/JSON, Kafka, PostgreSQL, Redis
Security: Keycloak (multi-tenant RBAC), Kong Gateway, HashiCorp Vault
Cloud: AWS (EKS, ECS, Lambda, RDS, ECR, ELB), Kubernetes
DeFi: Solidity, Move, Ethereum, BSC, Polygon, Aptos, Sui

WORK EXPERIENCE

Middle Software Engineer, Core Banking 03/2026 – Present

Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Developed and maintained the bank's **Lending** and **Overdraft** products as Python Smart Contracts on **Thought Machine Vault Core**.
- Implemented the **interest lifecycle** for both products: interest accrual plus penalty and late-fee calculation, scheduled through Vault Core events.
- Implemented **repayment and settlement** logic, the paths that move money and close out customer obligations.
- Integrated Vault Core with **external banking services** supporting core banking operations and transaction workflows.
- Built the **account migration system** (Golang on AWS ECS, Lambda, RDS), cutting migration time from **2.5 hours to 27 minutes (5x faster)**.
- Rebuilt the product **simulation testing framework** in Golang, so contract behaviour is verified in **just over 50 seconds instead of 24 minutes**.

Software Engineer

04/2025 – 03/2026

Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Built and operated AWS ETL pipelines for large-scale utility data (**150%** throughput gain) and migrated search from Elasticsearch to OpenSearch.

Software Engineer – Fintech Cloud Platform

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Implemented multi-tenant authentication and authorization with **Keycloak** and RBAC, isolating **20+ fintech and enterprise customer organizations**.
- Enforced service-level and network-level access control for **100+ microservices** through **Kong Gateway**.
- Operated **50+ production Kubernetes clusters** on OpenStack and Cluster API with zero-downtime upgrades.

Blockchain Developer – DeFi & Transaction Systems

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Integrated backend services with **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui), ensuring consistent **transaction handling and data integrity**.
- Developed and reviewed **15+ smart contracts** for DeFi and NFT use cases in Solidity and Move.
- Built event-driven services that synchronize on-chain events into off-chain databases and APIs.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built a gRPC-Gateway, Kafka, MongoDB, and Redis backend (**1.5K requests/second, 80–100 ms**) with multi-cluster failover for **99.95% uptime**.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera